Week 0 Technical Report: Infrastructure & Climate Data Analysis

Program: Kifiya AI Mastery (Week 0)

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Project: African Climate Trends Analysis (Sudan, Tanzania, Nigeria, Kenya, Ethiopia)

Objective: Establish a professional engineering environment and perform deep-dive data profiling on regional climate datasets to support African climate insights for COP32.

Part 1: Git & Environment Infrastructure (Task 1)

Before initiating data analysis, I established a robust development environment focused on version control, reproducibility, and automation. This foundational step ensures that all analytical work is traceable and can be replicated by other researchers.

1.1 Repository and Workspace Setup

The project infrastructure was built from the ground up to support collaborative data science:

- **Repository Initialization:** Created climate-challenge-week0 on GitHub.
- **Local Environment:** Initialized a local Python virtual environment (venv) to ensure that dependency versions remain constant, preventing "it works on my machine" issues.
- **Dependency Management:** All required libraries (Pandas, NumPy, Scipy, Matplotlib) were documented in a requirements.txt file.

1.2 Version Control Strategy

I implemented a professional feature-branching workflow. All infrastructure work was conducted on the setup-task branch before being integrated into the main codebase. I strictly followed **Conventional Commits** to maintain a meaningful project history:

- init: add .gitignore: Critical for security and repo health. I configured this to exclude data/ (preventing massive CSV uploads), .ipynb_checkpoints/, and local virtual environment folders.
- chore: venv setup: Standardized the execution environment.
- ci: add GitHub Actions workflow: Enabled automated validation of the setup.

1.3 Continuous Integration (CI)

I implemented a GitHub Actions pipeline (.github/workflows/ci.yml) to enforce high code quality. This pipeline triggers on every push to the main branch, performing the following automated steps:

1. **Runner Checkout:** Accessing the repository code.
2. **Python Setup:** Configuring the environment to Python 3.9/3.10.
3. **Automated Install:** Running pip install -r requirements.txt to verify that the environment can be built without errors.

Part 2: Data Preprocessing & Profiling Methodology (Task 2)

With the infrastructure secured, I performed a rigorous analysis of climate data for five African nations. This phase focused on converting raw, noisy satellite data into a clean, analysis-ready format.

2.1 Regional Branching & Loading

To prevent cross-contamination of data logic, I created dedicated branches for each country (e.g., eda-ethiopia, eda-sudan).

- **Loading:** Utilized pd.read_csv() for local data ingestion.
- **Country Attribution:** Injected a Country column into each DataFrame to allow for seamless merging during future multi-country comparisons.
- **Temporal Reconstruction:** NASA POWER data uses a YEAR and DOY (Day of Year) format. I used vectorized Pandas operations to synthesize a proper datetime index: pd.to_datetime(df["YEAR"] * 1000 + df["DOY"], format="%Y%j").
- **Feature Extraction:** Extracted Month to enable granular seasonal analysis, which is vital for identifying shifts in agricultural cycles across Africa.

2.2 Data Integrity and Quality Control

A significant portion of the work involved correcting data anomalies common in satellite-derived datasets:

- **Sentinel Mitigation:** Identified the -999 sentinel value (NASA's marker for missing data) and replaced it with np.nan. This was done *before* any statistical profiling to prevent massive negative skewing of temperature and precipitation averages.
- **Deduplication:** A rigorous scan using df.duplicated().sum() confirmed 0 redundant rows, verifying the chronological continuity of the source data.
- **Missing Value Strategy:** - Calculated null percentages per column; most variables showed high fidelity (%30 missing).
    - Applied **Forward-Fill (ffill)** to maintain temporal patterns in weather variables.
    - Established a "Quality Threshold": any row missing more than 30% of its parameters was dropped to ensure only high-confidence data points remained.

2.3 Statistical Outlier Detection (Z-Score Analysis)

I computed Z-scores for primary variables: T2M (Mean Temp), PRECTOTCORR (Precipitation), and WS2M (Wind Speed).

- **Detection:** Flagged all observations where $|Z| > 3$.
- **Reasoning & Retention:** After reviewing the flagged data, I chose to **retain** these outliers. In climate science, $|Z| > 3$ often represents critical extreme events—such as the 2015/2016 El Niño drought or flash flood events. Removing them would provide an inaccurately "stable" view of a rapidly changing climate.

Part 3: Detailed Country Analysis & Comparative Results

By reviewing the df.describe() and profiling outputs across the five notebooks, I identified distinct climatic signatures that are vital for the COP32 regional context.

3.1 Country-Specific Statistical Insights

Country	Pre-Processing Findings	Climate Signature Interpretation
Sudan	Highest T2M_MAX recorded ($>$). Low RH2M variance.	**Extreme Aridity:** The data reflects a desert climate where heatwaves are frequent. High Z-scores in temperature are common here.
Nigeria	Highest consistent RH2M (Humidity) and PRECTOTCORR.	**Tropical Stability:** The data shows a very predictable monsoon cycle with high moisture levels and lower diurnal temperature ranges.
Ethiopia	Lowest T2M_MIN ($<$ in highlands) and highest T2M_RANGE.	**Montane Variance:** High elevation results in significant daily temperature swings. Outlier detection here is sensitive to cold-fronts.
Kenya	Notable variability in WS2M (Wind Speed). Bimodal rain patterns.	**Equatorial Dynamic:** The analysis shows two distinct peak rainfall periods. Wind speed outliers indicate

		significant coastal/monsoonal influence.
Tanzania	High consistency in coastal regions. Significant RH2M.	**Coastal Tropical:** Similar to Nigeria but with a different bimodal rain cycle. Shows moderate temperature extremes compared to the Sahel.

3.2 Global Comparative Summary

1. **The North-South Thermal Gradient:** There is a stark contrast between the thermal maximums in Sudan and the more temperate, elevation-cooled climate of Ethiopia.
2. **Moisture Corridors:** Nigeria and Tanzania serve as the primary moisture basins in this study, with humidity levels often double those found in Sudan.
3. **Data Quality Consistency:** After the removal of -999 values, the datasets across all five nations showed high internal consistency, with T_{max} always correctly exceeding T_{min}, proving the effectiveness of the preprocessing steps.

4. Final Security & Export Standards

All finalized datasets were exported to data/<country>_clean.csv.

- **Verification:** I performed a final check of the .gitignore file to ensure the data/ folder is fully excluded. This maintains professional standards by keeping heavy data artifacts out of the version control system while sharing the high-quality analytical code.

End of Week 0 Technical Report